MCQ solutions

1. **B.** The first vector is twice the second. Moreover, their scalar product is not available in any basis; collinearity does not depend on a metric.
2. **A.** With the first two non-collinear vectors, linear combinations characterise their plane vector. The zero vector is a particular case, not the only one.
3. **C.** Non-collinear direction vectors rule out parallel and coincident lines. The intersection remains to be established: the lines may also be non-coplanar.
4. **B.** A basis guarantees the existence and uniqueness of the decomposition. The coefficients depend on the choice basis, which does not need to be orthonormal.
5. **C.** The dashed line keeps the edge and shows that it is hidden. Erasing the edge would remove information on the solid.
6. **B.** The directions AB, AC non-collinear generate the direction of the plane. A relation $AM = sAB + tAC$ therefore proves membership.
7. **B.** Two planes with the same direction are parallel in the broad sense. A point must be tested to distinguish coincident planes from strictly parallel ones.
8. **A.** We average each pair of coordinates: $(1;2;3)$. This midpoint formula does not does not even require orthonormality.
9. **C.** Perspective is a projection that loses depth. A drawn intersection does not prove not an intersection of lines in space.
10. **A.** Three non-coplanar vectors form a basis. Orthogonality and norm 1 are not required.

True/false. False: the lines (AB) and (CG) of a cube are non-coplanar. True: $\vec{0} = 0\vec{u}$. False: a projection can create apparent intersections; a calculation or geometric proof is necessary.

Solutions to the applications and guidance

Application 1. $\overrightarrow{JK} = 2\vec{u} - \vec{v} - \frac{1}{2}\vec{w}$; $\overrightarrow{IK} = \frac{3}{2}\vec{u} - \frac{3}{2}\vec{v}$.

Application 2. The first two drawings place the lines in a common plane; the third represent opposite edges of a non-degenerate tetrahedron. In a cube, (AB) and (CG) are not coplanar: the only plane containing A, B, C is that of the face $(ABCD)$, which does not contain G.

Application 3. (OA) and (BC) are neither parallel nor intersecting. Add "coplanar". Two lines distinct lines in the same plane, not parallel, intersect.

Application 4. The first coordinate forces any ratio between $\vec{v}$ and $\vec{u}$ i.e. 0, impossible since $\vec{v} \neq 0$. The combination is $(\lambda; \lambda + \mu; \mu) : \lambda = 2$, $\mu = 3$, so $m = 5$.

Application 5. An orthonormal coordinate system is needed. With two unit vectors at an angle of 60 °, the square of the norm of their sum is $1 + 1 + 2\cos 60° = 3$; the norm is $\sqrt{3}$. On the first pass, we can also see this length as a diagonal of a rhombus.

Application 6. $\overrightarrow{IJ} = \frac{1}{2}\overrightarrow{AB}$ and $\overrightarrow{IK} = \frac{1}{2}\overrightarrow{AC}$: two non-collinear directions of the planes are parallel, so the planes are parallel. If $O \in (IJK)$, then $\overrightarrow{IO} = \lambda\overrightarrow{IJ} + \mu\overrightarrow{IK}$. The coefficients of $\overrightarrow{OB}$ and $\overrightarrow{OC}$ require $\lambda = \mu = 0$, while that of $\overrightarrow{OA}$ would be $-\frac{1}{2} = 0$, contradiction.

Space research

Have students handle four non-coplanar points, then draw. In the correction, distinguish what what is a convention of representation and what is proved. To help, provide the three Chasles decompositions; to go further, replace the midpoints with the same ratio $t \in]0; 1[$ and recover the parallelism.

Teaching notes and solutions

Guidance for the teacher

Written test 1 comes before S12 and S15: set vector geometry in a given basis, without require new coordinate techniques. Use a physical cube, then a perspective drawing oblique (cavalier) projection; mark the hidden edges. Two non-parallel lines in space are not necessarily intersecting. The norm via coordinates requires an orthonormal coordinate system. The formal parametric representations and equations of planes will be dealt with later.

Solutions to the activity

1. By Chasles' relation, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = -\frac{1}{2}\vec{\imath} + \frac{1}{2}\vec{\jmath}$ and $\overrightarrow{IK} = -\frac{1}{2}\vec{\imath} + \frac{1}{2}\vec{k}$.

2. If $\overrightarrow{IK} = t\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would be $\frac{1}{2} = 0$, impossible in a basis. These two vectors are therefore non-collinear; I, J, K define a plane.

3. The centre of the cube satisfies $\overrightarrow{AO} = \frac{1}{2}(\vec{\imath} + \vec{\jmath} + \vec{k})$, therefore $\overrightarrow{IO} = \frac{1}{2}\vec{\jmath} + \frac{1}{2}\vec{k}$. If $O \in (IJK)$, there is s, t such that $\overrightarrow{IO} = s\overrightarrow{IJ} + t\overrightarrow{IK}$. Comparing the coefficients in the basis,

$$0 = -\frac{s+t}{2}, \qquad \frac{1}{2} = \frac{s}{2}, \qquad \frac{1}{2} = \frac{t}{2}.$$

The last two require $s = t = 1$, whereas the first would require $s + t = 0$. Contradiction : $O \notin (IJK)$.

4. Oblique (cavalier) perspective is a projection onto a sheet: distinct points may have the same image. It helps to conjecture, but an apparent overlap does not prove membership to a plane.

E1. The three coordinates of $\vec{v}$ are three times those of $\vec{u}$: $\vec{v} = 3\vec{u}$. The two vectors are therefore collinear.

E2. The midpoint has coordinates $(\frac{0+4}{2}, \frac{1-1}{2}, \frac{2+0}{2}) = (2, 0, 1)$. The difference of coordinates gives $\overrightarrow{AB} = (4, -2, -2)$; in the orthonormal coordinate system, $AB = \sqrt{16+4+4} = 2\sqrt{6}$.

E3. We calculate $2\vec{u} + 3\vec{v} = (2, 0, 2) + (0, 3, 3) = (2, 3, 5) = \vec{w}$. Thus $\vec{w}$ belongs to the vector plane spanned by the first two vectors: the three are coplanar.

E4. The first two vectors are non-collinear. If the third were a linear combination of them $\lambda(1, 0, 1) + \mu(0, 1, 1)$, the first two coordinates would require $\lambda = \mu = 1$, but the third would equal $2 \neq 0$. The three vectors are non-coplanar; they therefore form a basis of space.

E5. $\overrightarrow{AB} = (2, -1, 2)$ and $\overrightarrow{AC} = (4, -2, 4) = 2\overrightarrow{AB}$. Since $A \neq B$, the line (AB) is defined and this collinearity proves $C \in (AB)$.

Solutions to homework DM10

1. By Chasles' relation and the definitions of the midpoints,

$$\overrightarrow{IJ} = \frac{1}{2}\overrightarrow{AC} - \frac{1}{2}\overrightarrow{AB} = \frac{1}{2}\overrightarrow{BC}, \qquad \overrightarrow{IK} = \frac{1}{2}\overrightarrow{AD} - \frac{1}{2}\overrightarrow{AB} = \frac{1}{2}\overrightarrow{BD}.$$

2. The points B, C, D are not collinear, because the tetrahedron is not flat. The vectors $\overrightarrow{BC}, \overrightarrow{BD}$ are therefore non-collinear, as are $\overrightarrow{IJ}, \overrightarrow{IK}$. The planes have the same direction. They are distinct: if $I \in (BCD)$, the line $(BI) = (BA)$ would be contained in this plane, which would place A, contradiction. They are therefore strictly parallel.

3. In the basis $(\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD})$, we have $\overrightarrow{IG} = (-\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, $\overrightarrow{IJ} = (-\frac{1}{2}, \frac{1}{2}, 0)$ and $\overrightarrow{IK} = (-\frac{1}{2}, 0, \frac{1}{2})$. A relationship $\overrightarrow{IG} = s\overrightarrow{IJ} + t\overrightarrow{IK}$ imposes $s = t = \frac{1}{2}$ by the last two coordinates; the first would then be $-\frac{1}{2}$, and not $-\frac{1}{4}$. Hence $G \notin (IJK)$.

4. In this particular case, $I(2,0,0)$, $J(0,2,0)$ and $K(0,0,2)$. We find $IJ = \sqrt{(-2)^2 + 2^2} = 2\sqrt{2}$, and of same $IK = JK = 2\sqrt{2}$. The triangle is equilateral. This metric conclusion does not hold for every non-degenerate tetrahedron: it uses the coordinates of the particular case.

Solutions to the textbook or past-paper work

In this exercise, I is the midpoint of $[AB]$ and J that of $[CD]$.

1. The planes (AIE) and (BIG) both contain (AB). They are distinct: G does not belong to the plane of the face $ABFE$. Their intersection is therefore exactly the line (AB).

2. The points A, D, I are three non-collinear points of the face $ABCD$; the same holds for B, J, C. The two planes are therefore coincident with (ABC). The intersection is the whole plane, not a line.

3. The plane (HEF) is that of the top face; (BJC) is that of the bottom face. These planes are strictly parallel: their intersection is empty.